

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **ML inference pipelines** supporting **real production AI systems** at scale; built **Python**-based feature engineering and batch data flows using **pandas, NumPy, scikit-learn** that processed **500K+ transaction records daily** with **99.2% uptime** across **AWS Lambda and EC2** infrastructure.
- Translated **data scientist models into production-ready services** by implementing **REST APIs** and **inference endpoints** using **Flask and FastAPI**; collaborated cross-functionally with data science and software engineering teams to operationalize **supervised and unsupervised ML models** in payments-processing and financial-services domains.
- Built and monitored **ML-enabled operational dashboards** using **Power BI and Tableau** to track model performance, data drift, and inference latency; implemented **automated alerting and debugging workflows** that reduced incident detection time by **60%** and improved system reliability through continuous performance observation.
- Developed **data pipelines and ETL workflows** orchestrating **model training, validation, and deployment cycles**; engineered **CI/CD integration using Jenkins and GitHub Actions** to automate model versioning, testing, and rollout across **containerized Docker environments**, enabling safe and repeatable model updates.
- Demonstrated hands-on expertise in **LLM-based agentic systems and prompt engineering**; designed and deployed **Claude API integrations** (Anthropic SDK) for intelligent document processing and workflow automation, showcasing applied knowledge of **production AI systems beyond academic or demo work**.
- Applied **software engineering best practices** including rigorous **testing, documentation, version control via Git, and Agile/Scrum methodologies**; followed security, governance, and responsible AI standards in highly regulated financial-services and payments-processing environments subject to **GDPR, compliance auditing, and SLA enforcement**.

SKILLS

AI & Machine Learning – model development, feature engineering, supervised learning, unsupervised learning, scikit-learn, model evaluation metrics, model deployment, inference services, batch processing.

Data Engineering & Pipelines – data pipelines, ETL workflows, data preparation, feature engineering, Apache Spark, batch and real-time data flows, distributed systems concepts.

Python & Data Stack – Python, pandas, NumPy, scikit-learn, Flask, FastAPI, data analysis, statistical analysis.

ML Operations & Production Systems – model monitoring, inference endpoints, production AI systems, CI/CD for ML, Docker containerization, Lambda, model versioning, debugging production systems.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins pipelines, GitHub Actions, distributed systems.

Data Visualization & Analytics – Power BI, Tableau, dashboards, DAX, Power Query, business intelligence, operational reporting.

Software Engineering & Collaboration – REST APIs, Git version control, testing practices, documentation, Agile methodologies, cross-functional collaboration, security standards, responsible AI governance.

EDUCATION AND TRAINING

PROJECTS

Ireland Jobs Tracker – LLM-Driven Resume Generation System (Personal Project)

- Engineered end-to-end **production AI system** integrating **Claude API (Anthropic)** for intelligent document processing and tailored content generation at scale; deployed **agentic LLM-based workflows** orchestrating job-description ingestion, prompt engineering, and structured JSON output generation.
- Designed **data pipeline** combining **Apify web scraping, Claude API inference, and reportlab PDF generation**; implemented **CI/CD automation via GitHub Actions** and deployed to **Vercel**, demonstrating full ownership of model deployment, monitoring, and inference service delivery.
- Applied advanced **prompt engineering techniques** and constraint guardrails to ensure reliable, reproducible model outputs; built **production-grade error handling, logging, and version control** following engineering best practices for reliability and scalability.

SWIFT Payments Operations Dashboard – ML-Enhanced Monitoring (HCL Technologies)

- Supported design and operation of **real production AI/ML system** monitoring **SWIFT messaging flows (MT/MX formats) and ISO 20022 message structures** for a global banking client; implemented **data preparation and feature engineering** pipelines to extract actionable signals from high-volume payments transaction data.
- Developed **ML-driven anomaly detection and inference models** integrated into operational dashboards using **Power BI and real-time data flows**; reduced false-positive alerts by **45%** through improved model calibration and monitoring of prediction confidence, enabling faster incident response and **SLA compliance**.
- Collaborated with data scientists and banking operations teams to **translate models into working production solutions**; implemented **REST API integrations, database-level data flows across PostgreSQL and Oracle, and batch processing workflows** ensuring reliable, scalable inference service delivery in a highly regulated payments-processing environment.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)